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SIMULATION OF STATIC ROUTING PROTOCOL CONFIGURATION USING CISCO PACKET TRACER

AIM:

To simulate and configure the Static Routing Protocol using Cisco Packet Tracer and study how packets are transmitted between different networks using manually configured routes.

ALGORITHM:

1. Open Cisco Packet Tracer and create a new network topology.
2. Place two routers, two switches and four PCs in the workspace.
3. Connect two PCs to each switch and connect the switches to their respective routers.
4. Connect the two routers using an appropriate serial or Ethernet link.
5. Assign IP addresses and subnet masks to all PCs and router interfaces.
6. Configure the default gateway on each PC using the IP address of its router interface.
7. Configure the router interfaces and enable them using the no shutdown command.
8. Enter the static routing configuration on each router using the ip route command.
9. Verify the routing table using the show ip route command.
10. Use the ping command to test communication between PCs in different networks.
11. Observe packet transmission using Simulation mode in Cisco Packet Tracer.
12. Record the results and verify successful communication between the networks.

IP ADDRESS CONFIGURATION:

Network 1:

PC0 IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Router R1 LAN Interface: 192.168.1.1

Network 2:

PC1 IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Router R2 LAN Interface: 192.168.2.1

Router-to-Router Link:

R1 Interface: 10.0.0.1

R2 Interface: 10.0.0.2

Subnet Mask: 255.255.255.252

ROUTER R1 CONFIGURATION:

enable

configure terminal

ip route 192.168.2.0 255.255.255.0 10.0.0.2

end

ROUTER R2 CONFIGURATION:

enable

configure terminal

ip route 192.168.1.0 255.255.255.0 10.0.0.1

end

COMMANDS:

show ip route

ping 192.168.2.2

OUTPUT:

1. The routing table displays the manually configured static routes.
2. The ping command successfully transmits packets between PCs in different networks when all interfaces and routes are configured correctly.
3. The simulation displays packets passing through the routers to reach the destination network.

Sample output:

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=2ms TTL=126

Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

RESULT:

Thus, static routing was configured and simulated successfully using Cisco Packet Tracer. Static routes were added to the routing tables, and communication between different networks was verified using the ping command.